OptiscanAI: A Benchmark and Deployment Study of Large Vision Models for Multi-Disease Retinal Screening

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Abstract—**Retinal disease is a leading cause of preventable blindness, and specialist capacity in sub-Saharan Africa cannot meet demand. This paper reports what a modular multi-disease screening system built on a retinal foundation model does and does not deliver, evaluated on the held-out official test split of the Retinal Fundus Multi-Disease Image Dataset. We report per-disease sensitivity, specificity, positive predictive value, false-negative rate and area under the precision-recall curve with bootstrap confidence intervals, a controlled ablation over the backbone and adaptation strategy, and a direct comparison of full and 8-bit precision. Three findings matter for deployment. First, discrimination is strong but heavily disease-dependent: macro AUC is 0.872, ranging from 0.996 for myopia to 0.663 for optic disc pallor. Second, the model is severely over-confident, with an expected calibration error of 0.209; this makes the shipped threshold policy degenerate, silencing five classes entirely, and a rank-preserving recalibration reduces the error to 0.003. Third, as a referral triage the system reaches an AUC of 0.952, but transferring that operating point to primary-care prevalence drops positive predictive value from 0.93 to below 0.30. We therefore present these results as benchmark evidence and a deployment-constraint analysis, not as clinical validation in Uganda.**

Index Terms—**retinal screening, multi-label classification, model calibration, explainability, low-resource deployment, foundation models**

I. Introduction

Retinal disorders are among the leading and fastest-growing causes of avoidable blindness. The World Health Organization estimates that at least 2.2 billion people live with vision impairment, much of it treatable if detected early [1]. Diabetic retinopathy alone affected more than 103 million people in 2020 [2]. In Uganda, specialist eye care is concentrated in urban centres, so primary health providers are the first and often only point of contact for ocular disease, and referral decisions are made by clinicians who are not ophthalmologists.

Deep learning on fundus photographs has reached expert-level performance on individual diseases [3], and domain-specific foundation models now transfer across ophthalmic tasks [8]. Translating that into benefit in a resource-constrained clinic is a different problem: it requires handling severe class imbalance, producing predictions a non-specialist can act on, operating without reliable connectivity, and surviving distribution shift [4], [5].

An earlier version of this work described the OptiscanAI architecture and reported aggregate benchmark scores. Reviewers correctly observed that aggregate scores on a public Indian-sourced corpus cannot support claims about Ugandan clinical performance, and that the contribution of each component was not separable. This paper is the response. We keep the system description but re-position the empirical claim: what follows is benchmark evidence and a deployment-constraint analysis, with the gap to clinical readiness stated quantitatively rather than as a caveat.

Contributions. (i) A per-disease evaluation on the held-out official RFMiD test split reporting sensitivity, specificity, PPV, FNR and AUPRC with 95% bootstrap confidence intervals, rather than aggregate F1 alone. (ii) A controlled ablation isolating retinal pretraining, low-rank adaptation, the decision-threshold policy and 8-bit quantisation under a shared training recipe. (iii) The identification of a calibration failure that silently disables five disease channels in the shipped configuration, and a rank-preserving correction that removes it. (iv) A quantitative faithfulness test of the explainability layer, both the saliency maps and the generated text. (v) A prevalence-transfer analysis that converts the benchmark operating point into the referral burden a Ugandan health centre would actually see.

II. Related Work

A. Retinal Detection and Foundation Models

Convolutional networks for retinal image analysis were established by Gulshan *et al.* [3] for diabetic retinopathy, and later extended to glaucoma and age-related macular degeneration, predominantly one disease at a time. Multi-label classification across a broad taxonomy, which is what primary-care triage requires, remains hard because of extreme class imbalance, label co-occurrence and the scarcity of positives for rare conditions [6]. Vision Transformers [7] capture the long-range spatial dependencies relevant to retinal pathology, and RETFound [8], pretrained by masked autoencoding on 1.6 million colour fundus photographs, improves downstream ophthalmic tasks over ImageNet initialisation. Low-Rank Adaptation [9] adapts such a backbone while training a small fraction of its parameters, which matters when the institutional compute budget is a single GPU.

B. Imbalance, Calibration and Explainability

Asymmetric losses [15] and focal losses [14] address imbalance by down-weighting easy negatives. A known side effect is distorted probability estimates, and modern networks are over-confident even without such losses [16]; recalibration by Platt scaling or temperature scaling restores usable probabilities without changing the ranking. Grad-CAM [10] and Shapley-based attributions [11], [12] are the standard post-hoc explanations, and the deletion/insertion protocol [13] is the standard way to test whether such maps reflect the evidence the model actually uses. Real-world deployment studies show that workflow fit, image quality and clinician trust dominate practical effectiveness [4].

III. System Architecture

Fig. 1 shows the pipeline; the numbers below refer to its labelled components.

A. Input Quality Gating (1)

A four-layer gate rejects unusable images before inference: structural checks on resolution and aspect ratio; statistics for illumination uniformity, focus sharpness, field-of-view adequacy and circular aperture; a learned MobileNetV3-Small binary classifier; and a fusion rule that accepts an image when $0.6\, s_{\text{stat}} + 0.4\, s_{\text{learned}} \geq 0.70$. Rejected images trigger recapture guidance rather than a prediction.

B. Backbone and Adaptation (2)–(3)

The backbone is RETFound ViT-L/16 [8] operating at 224×224, giving 196 patch tokens and a 1024-dimensional representation that is projected to 512. LoRA adapters of rank $r{=}16$ with $\alpha{=}32$ are injected into the query-key-value projection of each of the 24 transformer blocks, so

$$W' = W_0 + \frac{\alpha}{r} BA, \tag{1}$$

where W_0 is frozen and only $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ are trained. This leaves 2.44 M trainable parameters out of 305.7 M, or 0.8%. A bottleneck head ($512 \to 128 \to 24$, dropout 0.5 and 0.3) produces the multi-label output. Within the network, a single sparse top-k attention layer ($k{=}32$) over the patch tokens provides spatial context pooling before classification; we note explicitly that this is patch-level self-attention, *not* a graph attention network over the clinical knowledge graph. Classes with fewer than 10 training positives are dropped, reducing the RFMiD taxonomy from 45 labels to the 24 that can be learned at all.

C. Loss and Decision-Threshold Policy (4)

Screening tolerates false alarms far better than misses, but an unusable alarm rate destroys the referral pathway. Training uses an asymmetric loss [15] that never down-weights positives and aggressively suppresses easy negatives, with $\gamma_{\text{pos}}{=}0$, $\gamma_{\text{neg}}{=}4$, probability clipping at 0.05 and label smoothing 0.05. After training, a per-class threshold τ_c is fitted on the validation split as the lowest threshold whose precision still satisfies a floor of 0.10; classes that cannot meet the floor at any threshold are assigned $\tau_c{=}0.95$, which in practice disables them. Section V shows that this policy interacts badly with the model's calibration, and reports the correction.

D. Clinical Knowledge Graph (5)

The knowledge graph is a curated symbolic layer, not a learned module, and it is applied *after* classification. For the 24 deployed classes it holds 24 disease nodes and

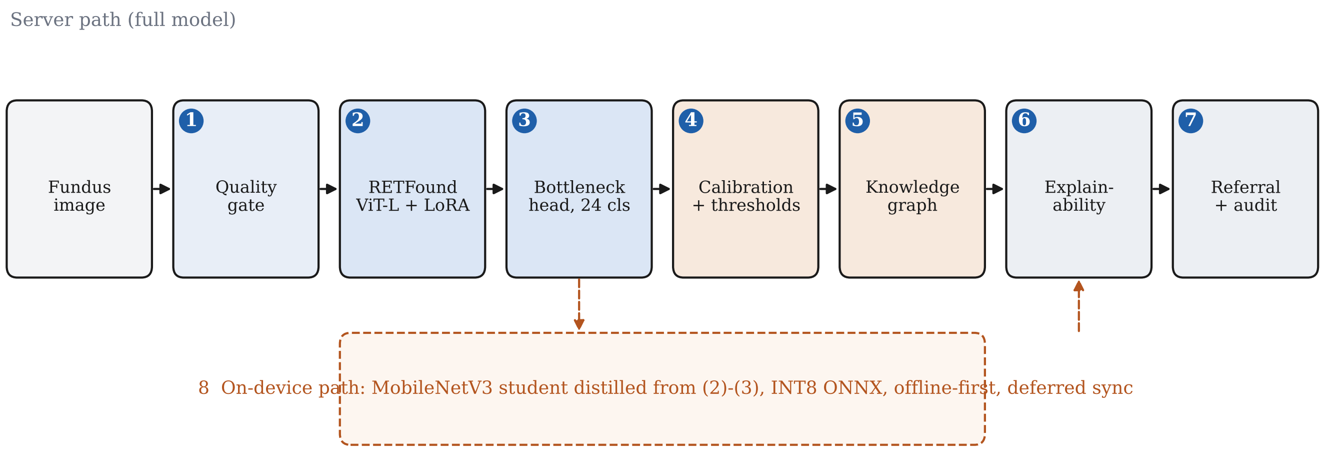

Fig. 1. OptiscanAI pipeline. (1) image quality gate; (2) RETFound ViT-L/16 backbone with LoRA adapters; (3) bottleneck classification head over 24 retained classes; (4) probability calibration and per-class decision thresholds; (5) clinical knowledge graph applied post-hoc; (6) explainability layer; (7) referral decision with an audit record. Component (8) is the offline path: a MobileNetV3 student distilled from (2)–(3), exported to INT8 ONNX, which runs on-device and defers synchronisation until connectivity returns.

54 typed edges over six relation families: co-occurrence, category membership (vascular, degenerative, glaucomatous, infectious/immunologic, neuro-ophthalmic, tractional), severity grade on a three-point scale, systemic association with weights, age-related risk, and treatment consideration. Content was extracted by the authors from clinical practice guidelines and primary literature, with the source recorded per disease: the American Academy of Ophthalmology Preferred Practice Pattern for primary open-angle glaucoma (2022) for optic disc cupping, AREDS Report No. 39 for age-related macular degeneration, and the Uganda Ministry of Health diabetes guidelines (2024) for diabetic retinopathy, with per-condition journal references for vein occlusion and retinal pigment epithelial changes carried in the released graph file. Uganda-specific prevalence weights are attached to six conditions. The graph has *not* been formally validated by an independent ophthalmologist panel; that review is outstanding and we do not claim it. Its function at inference is to refine co-occurring probabilities, assign a three-level referral priority and compute a composite risk score; Section V measures what that is worth.

E. Explainability (6) and Offline Deployment (8)

The explainability layer combines Grad-CAM [10] attention maps, Integrated Gradients [12] attributions, and short natural-language reports produced by a distilled narrator constrained to the knowledge-graph vocabulary. For deployment, a MobileNetV3-Large student (5.2 M parameters) is distilled from the teacher with a four-term objective combining soft-target, task, feature and threshold-consistency losses, and exported to INT8 ONNX at 5.4 MB from a 20.8 MB FP32 export. On-device inference removes the cloud dependency, and records synchronise when connectivity returns. All clinical interactions are logged for audit.

IV. Experimental Setup

Data. We use the Retinal Fundus Multi-Disease Image Dataset [6] with its official partition: 1,920 training, 640 validation and 640 test images, annotated across 45 disease categories with prevalence from below 1% to above 50%. All results below are on the 640-image test split, which is used for nothing else: model selection and every threshold are fitted on the validation split.

Preprocessing. Images are resized directly to 224×224 with bicubic interpolation and normalised with ImageNet statistics. Two reviewer questions concern this choice, and rather than argue them we measured both as ablation arms in Table II. ImageNet statistics are retained because the backbone was pretrained under them: matching the *pretraining* input distribution matters more than matching the fine-tuning corpus, since the LoRA adapters modify a representation learned in that space. Substituting RFMiD's own channel statistics costs 0.054 AUC and 0.107 AUPRC, which supports keeping them. We also confirm that the deployed pipeline applies *no* histogram equalisation; an earlier draft described one, and it is not in the implementation. Since the reviewer's concern was that equalisation may distort colour-dependent findings, we tested CLAHE on the L channel of LAB rather than assume either way, and report the result in Section V-B.

Training. The deployed checkpoint uses AdamW with a head learning rate of 5×10^{-4}, backbone 10^{-6} after staged unfreezing of the last four blocks at epoch 10, weight decay 10^{-2}, batch size 16, cosine annealing, 25 epochs with early stopping on validation macro precision, bf16 mixed precision, and six-view test-time augmentation. Every ablation arm in Table II instead shares one common recipe – 15 epochs, batch 32, cosine annealing, identical augmentation, selection on validation mAP – so that arms differ only in backbone and adaptation.

Metrics. We report AUC-ROC and AUPRC, which are threshold-free, alongside sensitivity, specificity, PPV, FNR and FPR at explicit operating points. Confidence intervals are 95% percentile bootstrap over 1,000 resamples of the test images. Calibration is measured as expected calibration error over 15 equal-width bins.

Reporting standards. The revision follows CLAIM 2024 [17] and TRIPOD+AI [18]: one held-out split used once, per-outcome discrimination *and* calibration, interval estimates on headline figures, and an explicit intended population. Against the six FUTURE-AI principles [19], this study evidences traceability, explainability (Section V-F, including where it fails) and robustness under quantisation (Section V-E), and evidences neither fairness nor universality, which need the demographically annotated local data we do not have.

V. Results

A. Per-Disease Performance

Table I gives the full per-disease breakdown. Macro AUC is 0.872 (95% CI 0.846–0.896) and macro AUPRC is 0.357 (95% CI 0.345–0.406). The aggregate hides a wide spread that matters clinically: myopia reaches an AUC of 0.996, media haze 0.974, age-related macular degeneration 0.972 and diabetic retinopathy 0.967 with an AUPRC of 0.873, while optic disc pallor (0.663), macular hole (0.674) and asteroid hyalosis (0.687) are close to uninformative. Fig. 2(b) shows the pattern: performance tracks the number of positive examples, and every class below roughly 20 test positives has a confidence interval too wide to support a clinical claim.

B. Backbone and Adaptation Ablation

Table II isolates the components. Retinal pretraining carries most of the signal: freezing RETFound and training only the head reaches AUC 0.877 while updating 0.8% of the parameters, already ahead of every ImageNet baseline. Low-rank adaptation then adds a real but modest increment, taking AUC to 0.882 and AUPRC from 0.360 to 0.392 for 1.6 M extra trainable parameters, so LoRA is worth its cost. The comparison that qualifies the story is ViT-B/16: fine-tuned end-to-end on ImageNet initialisation it reaches a lower AUC of 0.871 but a substantially higher AUPRC of 0.472, the best of any arm. Because AUPRC is the metric that reflects performance under class imbalance, the fair conclusion is not that the retinal foundation model dominates. It gives the best ranking while training 1.3% of its parameters, which is what makes single-GPU institutional training feasible, but a fully fine-tuned ImageNet transformer retrieves rare positives better. We report this because the previous draft implied a larger and less qualified benefit than the evidence supports.

The same table settles the two preprocessing questions empirically. Replacing ImageNet normalisation with RFMiD's own channel statistics costs 0.054 AUC and 0.107 AUPRC, confirming that matching the pretraining distribution matters more than matching the fine-tuning corpus. Adding CLAHE costs 0.016 AUC and 0.051 AUPRC. The second result supports the reviewer's concern directly: contrast equalisation is not free on fundus images, and on this corpus it removes more diagnostic signal than it recovers, which is a reason to leave it out rather than an oversight. The post-hoc rows complete the picture. Quantisation is discussed in Section V-E; the knowledge graph moves macro AUC by -0.0002 because its reasoning rules alter only 12.7% of images and a single class, for the reason given in Section V-G.

C. What the Threshold Policy Costs

Table III answers the reviewer question about the other side of the trade-off. The deployed precision-floor policy is not, as previously described, a false-positive reduction: it is a recall-first policy. Relative to a uniform $\tau{=}0.5$ it reduces false negatives from 199 to 81 across the test split, a 59% reduction, but raises false positives from 484 to 3,858 and drops macro PPV from 0.383 to 0.114, taking the alert rate from 1.5 to 6.9 findings per image. It also leaves five classes permanently silent. A policy that instead fixes a sensitivity target of 0.90 on validation dominates it on the test split: macro sensitivity 0.811 against 0.591, macro PPV 0.189 against 0.114, fewer false negatives, and no silent classes.

D. Calibration

The behaviour above has a single cause. The model's probabilities are severely over-confident: mean predicted probability is 0.251 against an empirical positive rate of

TABLE I

Per-disease performance on the held-out official RFMiD test split (n=640 images, 24 retained classes). AUC and AUPRC are threshold-free; the operating-point columns use the per-class thresholds τ_c fitted on the validation split under the deployed precision-floor policy. Brackets give 95% percentile bootstrap confidence intervals over 1,000 image resamples. FNR is the false-negative rate and the false-positive rate is its complement on the negative side, $\mathrm{FPR} = 1 - \mathrm{Spec.}$; a dash marks a quantity that is undefined because the class never fires.

Class	n_+	Prev. (%)	AUC	95% CI	AUPRC	τ_c	Sens.	Spec.	PPV	FNR
DR	124	19.4	0.967	[0.95, 0.98]	0.873	0.05	1.000	0.000	0.194	0.000
ARMD	31	4.8	0.972	[0.95, 0.99]	0.695	0.17	1.000	0.575	0.107	0.000
MH	104	16.2	0.974	[0.96, 0.98]	0.875	0.05	1.000	0.000	0.163	0.000
DN	46	7.2	0.792	[0.73, 0.85]	0.203	0.27	0.957	0.316	0.098	0.043
MYA	32	5.0	0.996	[0.99, 1.00]	0.931	0.13	1.000	0.528	0.100	0.000
BRVO	23	3.6	0.919	[0.85, 0.97]	0.649	0.29	0.826	0.793	0.129	0.174
TSLN	53	8.3	0.940	[0.91, 0.97]	0.638	0.05	1.000	0.000	0.083	0.000
ERM	5	0.8	0.787	[0.59, 0.95]	0.049	0.37	0.200	0.986	0.100	0.800
LS	15	2.3	0.868	[0.74, 0.96]	0.189	0.31	0.800	0.811	0.092	0.200
MS	7	1.1	0.903	[0.75, 0.98]	0.147	0.37	0.714	0.942	0.119	0.286
CSR	13	2.0	0.907	[0.81, 0.97]	0.314	0.37	0.615	0.931	0.157	0.385
ODC	91	14.2	0.783	[0.73, 0.84]	0.530	0.05	1.000	0.000	0.142	0.000
CRVO	9	1.4	0.961	[0.92, 1.00]	0.578	0.27	0.778	0.919	0.121	0.222
AH	5	0.8	0.687	[0.48, 0.90]	0.029	0.95	0.000	1.000	–	1.000
ODP	24	3.8	0.663	[0.59, 0.74]	0.054	0.95	0.000	1.000	–	1.000
ODE	17	2.7	0.922	[0.82, 1.00]	0.662	0.23	0.824	0.761	0.086	0.176
AION	4	0.6	0.702	[0.38, 0.95]	0.093	0.35	0.250	0.951	0.031	0.750
PT	4	0.6	0.930	[0.86, 0.98]	0.067	0.95	0.000	1.000	–	1.000
RT	5	0.8	0.831	[0.71, 0.96]	0.039	0.35	0.200	0.964	0.042	0.800
RS	14	2.2	0.981	[0.96, 1.00]	0.619	0.19	1.000	0.859	0.137	0.000
CRS	11	1.7	0.907	[0.77, 0.98]	0.182	0.57	0.273	0.976	0.167	0.727
EDN	4	0.6	0.914	[0.85, 0.97]	0.050	0.95	0.000	1.000	–	1.000
RPEC	4	0.6	0.941	[0.86, 0.98]	0.081	0.41	0.750	0.959	0.103	0.250
MHL	3	0.5	0.674	[0.33, 0.95]	0.016	0.95	0.000	1.000	–	1.000
Macro	648	–	0.872	[0.85, 0.90]	0.357	–	0.591	0.720	0.114	0.409

RFMiD label abbreviations: DR, diabetic retinopathy; ARMD, age-related macular degeneration; MH, media haze; DN, drusen; MYA, myopia; BRVO, branch retinal vein occlusion; TSLN, tessellation; ERM, epiretinal membrane; LS, laser scar; MS, macular scar; CSR, central serous retinopathy; ODC, optic disc cupping; CRVO, central retinal vein occlusion; AH, asteroid hyalosis; ODP, optic disc pallor; ODE, optic disc edema; AION, anterior ischaemic optic neuropathy; PT, parafoveal telangiectasia; RT, retinal traction; RS, retinitis; CRS, chorioretinitis; EDN, exudation; RPEC, retinal pigment epithelium changes; MHL, macular hole.

0.042, the minimum probability over the entire test split is 0.064, and expected calibration error is 0.209. Because nothing is ever confidently negative, the precision-floor search drives thresholds to the bottom of the grid for high-prevalence classes, where specificity collapses to zero, and to the fallback for rare ones, where sensitivity collapses to zero. Per-class Platt scaling fitted on the validation split reduces test ECE to 0.003 (macro ECE 0.210 to 0.013) and, being monotone, leaves AUC unchanged at 0.872. Fig. 2(a) shows the before and after. Applied together with the threshold policy, recalibration cuts the false-positive count from 3,986 to 2,356 and raises macro PPV from 0.189 to 0.216. We attribute the miscalibration to the asymmetric loss: probability clipping at 0.05 removes the gradient that would push confident negatives towards zero, and label smoothing bounds the targets away from the extremes.

E. Effect of 8-Bit Quantisation

Table IV compares full and 8-bit precision directly on the same 640 test images with identical CPU threading, replacing the previous unsupported claim of negligible loss. The claim holds for ranking and fails for retrieval: macro AUC is unchanged at 0.873 against 0.872, but macro AUPRC falls from 0.357 to 0.346, a 3.1% relative loss concentrated in the rare classes whose positive examples are already scarce. At the deployed operating point the quantised model is slightly more permissive, catching eight more true positives at the cost of 218 additional false alarms. Two practical points follow. First, a screening system should be re-thresholded after quantisation rather than inheriting the full-precision thresholds, because quantisation shifts the operating point even when it preserves the ranking. Second, dynamic quantisation is a storage optimisation, not a speed one for this architecture: it cuts the exported model by 74.7% but reduces CPU latency by only 4%, because PyTorch dynamic quantisation compresses linear weights while the attention matrix products that dominate a ViT-L forward pass stay in floating point. The latency benefit in deployment comes instead from the distilled MobileNetV3 student, which runs in 102 ms per image at 5.4 MB and agrees with its own FP32 export

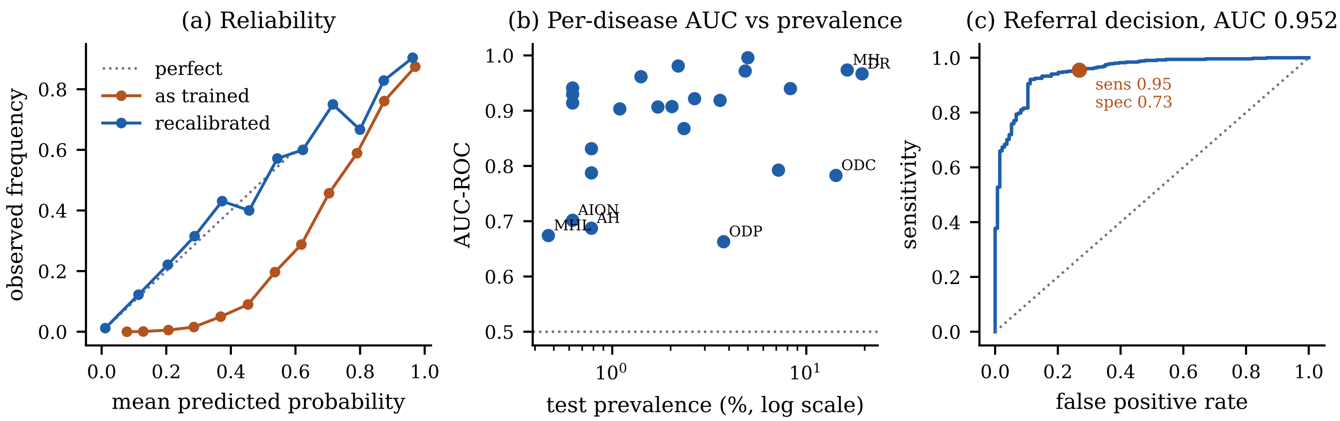

Fig. 2. (a) Reliability on the test split before and after per-class Platt scaling; the as-trained curve sits far below the diagonal, which is over-confidence. (b) Per-disease AUC against test prevalence, log scale; the weakest classes are the rarest. (c) ROC for the referral decision, with the validation-fitted high-sensitivity operating point marked.

TABLE II

COMPONENT ABLATION ON THE RFMID TEST SPLIT. THE ADAPTATION ARMS SHARE THE DATA, THE ASYMMETRIC LOSS, A 15-EPOCH SCHEDULE, SELECTION ON VALIDATION MAP AND THE SAME THRESHOLD OPTIMISATION, SO DIFFERENCES ARE ATTRIBUTABLE TO THE COMPONENT NAMED. POST-HOC ROWS ARE APPLIED TO THE DEPLOYED CHECKPOINT, SO THEY ARE EXACT RATHER THAN RE-TRAINED. THE GRAPH ALTERS 12.7% OF IMAGES AND ONLY ONE CLASS (ODE), BECAUSE THE PARTNER CONDITIONS ITS CO-OCCURRENCE RULES ENCODE ARE MOSTLY AMONG THE 21 CLASSES DROPPED FOR HAVING TOO FEW TRAINING POSITIVES. †THE DEPLOYED CHECKPOINT ADDITIONALLY USES 25 EPOCHS, STAGED BACKBONE UNFREEZING AND SIX-VIEW TEST-TIME AUGMENTATION.

Arm or component	Params (M)	Train. (M)	% train.	AUC	AUPRC
Backbone and adaptation (shared 15-epoch recipe)					
ResNet-50 (ImageNet, full FT)	23.6	23.56	100.0	0.867	0.336
EfficientNet-B3 (ImageNet, FT)	10.7	10.73	100.0	0.819	0.377
ViT-B/16 (ImageNet, full FT)	85.8	85.82	100.0	0.871	0.472
RETFound ViT-L, head only	305.7	2.44	0.8	0.877	0.360
RETFound ViT-L + LoRA r=16	307.3	4.01	1.3	0.882	0.392
Input preprocessing (RETFound + LoRA arm)					
RFMiD-specific normalisation	–	–	–	0.827	0.285
ImageNet norm. + CLAHE	–	–	–	0.866	0.341
Post-hoc components (deployed checkpoint)					
+ knowledge-graph reasoning	–	–	–	0.871	0.357
+ INT8 dynamic quantisation	–	–	–	0.873	0.346
Deployed checkpoint†	305.7	2.44	0.8	0.872	0.357

TABLE III

WHAT THE THRESHOLD POLICY COSTS ON EACH SIDE OF THE TRADE-OFF, ON THE RFMID TEST SPLIT. COUNTS ARE SUMMED OVER ALL 24 CLASSES AND 640 IMAGES. "SILENT" COUNTS CLASSES WHOSE THRESHOLD IS SO HIGH THAT THE CLASS NEVER FIRES, SO ITS SENSITIVITY IS IDENTICALLY ZERO.

Policy	Sens.	Spec.	PPV	FN	FP	Silent
As trained						
Uniform τ=0.5	0.420	0.965	0.383	199	484	6
Precision floor (deployed)	0.591	0.720	0.114	81	3858	5
Sensitivity target $\geq$ 0.90	0.811	0.729	0.189	75	3986	0
After per-class Platt recalibration						
Uniform τ=0.5	0.244	0.994	0.733	348	84	13
Precision floor (deployed)	0.347	0.943	0.439	194	764	11
Sensitivity target $\geq$ 0.90	0.657	0.835	0.216	92	2356	1

TABLE IV

EFFECT OF 8-BIT DYNAMIC QUANTISATION ON THE REFERENCE MODEL, ON THE SAME 640 TEST IMAGES WITH IDENTICAL CPU THREADING, SO ACCURACY, SIZE AND LATENCY ARE DIRECTLY COMPARABLE. THE DEPLOYED ON-DEVICE STUDENT COMPRESSES FURTHER, FROM 20.8 MB TO 5.4 MB ONNX (−73.8%), AGREEING WITH ITS FP32 EXPORT ON 99.6% OF BINARY DECISIONS AT τ=0.5 AND RUNNING AT 102 MS MEDIAN PER IMAGE.

Precision	MB	AUC	AUPRC	Sens.	PPV	ms/img
FP32	1224	0.872	0.357	0.591	0.114	434
INT8 dynamic	310	0.873	0.346	0.610	0.106	418
Change	−74.7%	+0.002	-0.011	+0.018	-0.008	−4%

on 99.6% of binary decisions.

F. Faithfulness of the Explanations

Table V reports the deletion/insertion test [13] on 60 confidently detected true positives, against a random-saliency control on the same images. Both attribution methods pass the deletion test: removing the regions they highlight degrades the target probability significantly faster than removing random regions (Grad-CAM p=0.029, Integrated Gradients p=0.008, one-sided Wilcoxon). Neither passes the insertion test, where both are statistically indistinguishable from random. The maps therefore identify regions the model uses, but they do not identify a region set sufficient to reproduce the prediction, and the flat insertion curves are a direct

TABLE V
FAITHFULNESS OF THE EXPLAINABILITY LAYER ON 60 CONFIDENTLY DETECTED TRUE POSITIVES. DELETION AUC IS BETTER WHEN LOWER, INSERTION AUC WHEN HIGHER; p-VALUES ARE ONE-SIDED WILCOXON SIGNED-RANK TESTS AGAINST THE RANDOM-SALIENCY CONTROL ON THE IDENTICAL IMAGES. NARRATIVE LAYER (n=24 REPORTS FROM THE DISTILLED NARRATOR THAT ALWAYS GENERATES AND NEVER OMITS A DETECTED FINDING): PROBABILITY INFIDELITY 0% (95% CI 0%–14%), FINDING OMISSION 0%, UNSUPPORTED SEVERITY ESCALATION 21% (95% CI 9%–40%).

Attribution	Del. AUC	vs. rand.	Ins. AUC	vs. rand.
Grad-CAM	0.913	$p = 0.029$	0.926	$p = 0.996$
Integrated Grad.	0.861	$p = 0.008$	0.943	$p = 0.590$
Random (control)	0.946	–	0.943	–

consequence of the calibration problem: predicted probabilities never fall below 0.064, so revealing evidence has little room to move them. We accordingly present the maps as attention hints requiring clinician verification, not as standalone evidence. The narrative layer is audited separately for probability infidelity, omission of detected findings, and unsupported escalation of urgency.

G. Referral Triage and What the Knowledge Graph Adds

Per-disease multi-label metrics understate the system, because the output a primary-care operator acts on is a single referral decision. Scored against the RFMiD `Disease_Risk` flag, the maximum class probability reaches an AUC of 0.952 (95% CI 0.932–0.968) and an AUPRC of 0.986. At the validation-fitted high-sensitivity operating point, test sensitivity is 0.955 and specificity 0.731, missing 23 of 506 positive images.

The knowledge graph's measurable contribution is here, not in classification accuracy. Its co-occurrence rules almost never fire on the retained label set, because most of the partner conditions they encode are among the 21 classes dropped for having too few training examples; claiming a macro-F1 improvement from the graph, as the previous draft did, is not supportable. Its referral-priority rules, however, do stratify risk monotonically once the thresholds are non-degenerate: under the sensitivity-targeted policy the graph routes 84.5% of images to URGENT (86.9% of which carry pathology), 10.8% to ROUTINE (47.8%), and 4.7% to FOLLOW-UP (10.0%). Under the shipped thresholds the same rules classify 100% of images as URGENT and carry no information at all, which is a further consequence of the calibration failure.

TABLE VI
WHY A BENCHMARK RESULT IS NOT A CLINICAL RESULT. THE REFERRAL OPERATING POINT IS HELD FIXED AT THE MEASURED TEST SENSITIVITY 0.955 AND SPECIFICITY 0.731; ONLY THE PREVALENCE OF THE SCREENED POPULATION CHANGES. RFMID IS AN ENRICHED RESEARCH CORPUS, SO ITS POSITIVE RATE IS FAR ABOVE ANY PRIMARY-CARE CASELOAD.

Population	Prev. (%)	PPV	NPV	Refer (%)
RFMiD test (as measured)	79	0.931	0.810	81
Projected at 30% prev.	30	0.604	0.974	47
Projected at 20% prev.	20	0.470	0.985	41
Projected at 10% prev.	10	0.283	0.993	34
Projected at 5% prev.	5	0.158	0.997	30

H. From Benchmark to Clinic

Table VI is the reason this paper does not claim clinical readiness. RFMiD is an enriched research corpus in which 79% of test images carry pathology. Holding the measured sensitivity and specificity fixed and varying only prevalence, PPV falls from 0.931 on the benchmark to 0.470 at 20% prevalence and 0.158 at 5%, while roughly 30% of all screened patients would still be referred. A screening programme sized on the benchmark PPV would therefore under-provision its referral pathway by a factor of three or more. Negative predictive value moves the other way, exceeding 0.99 at low prevalence, which is the property a rule-out triage tool most needs.

VI. DISCUSSION AND LIMITATIONS

The benchmark results here do not constitute evidence of clinical effectiveness in Uganda, and we state the gap concretely rather than as a disclaimer. RFMiD is sourced predominantly from Indian populations and acquired on clinic-grade cameras. No Ugandan fundus images were used at any stage of this study, so nothing here speaks to performance under Ugandan retinal pigmentation, comorbidity patterns such as HIV-associated retinopathy, or the handheld and smartphone-adapted cameras that a district health centre would realistically use. The prevalence analysis above quantifies one part of that gap; distribution shift in image appearance is a second part we have not measured at all.

Three further limitations bound the claims. Rare classes are unreliable: 21 of the 45 RFMiD labels were dropped for having fewer than 10 training positives, and among the 24 retained, those with fewer than 20 test positives have intervals wide enough to include chance. The knowledge graph has documented sources but no independent ophthalmologist review. And the ablation arms share a shortened 15-epoch recipe, so they bound each component's relative contribution rather than its best achievable accuracy.

The path to a defensible clinical claim is prospective evaluation on locally collected images against a reference standard read by Ugandan ophthalmologists, with the operating point chosen for the local prevalence rather than the benchmark's. Regulatory approval from the Uganda National Drug Authority, compliance with the Data Protection and Privacy Act (2019), ISO 13485 quality management and adherence to Good Machine Learning Practice principles remain prerequisites for any deployment. A working prototype of the interface described in Section III is available at https://landwind22-retinal-screening.hf.space; it is a demonstration of the workflow, not a cleared medical device.

VII. Conclusion

We re-evaluated OptiscanAI on the held-out RFMiD test split and report what the evidence supports. Discrimination is strong in aggregate (macro AUC 0.872), excellent for the common sight-threatening conditions that dominate primary-care caseloads, and reaches an AUC of 0.952 as a referral triage. Against that, the ablation shows the retinal backbone with LoRA gives the best ranking while training 1.3% of its parameters yet is still beaten on average precision by a fully fine-tuned ImageNet transformer; the shipped threshold policy is recall-first rather than precision-improving and silently disables five disease channels; and the probabilities are badly calibrated, which is the common cause of the degenerate thresholds, the uninformative referral prioritisation and the flat insertion curves in the explainability audit. A rank-preserving recalibration removes that failure mode. Transferring the benchmark operating point to primary-care prevalence cuts positive predictive value by more than half, so we position this work as benchmark evidence and a deployment-constraint study, with prospective validation on Ugandan data the outstanding next step.

Acknowledgments

The authors thank the College of Computing and Information Sciences, Makerere University, for advice and consultations during this work, and acknowledge the creators of the RFMiD dataset for making the data publicly available. We thank the anonymous reviewers, whose questions about per-disease evidence, component attribution and the Uganda framing directly shaped this revision.

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